

# SQL Capstone Project: Zomato Database Design

## Step 1: Choose a Leading Platform

Platform: Zomato (Food Delivery & Restaurant Discovery Platform)

Industry: Food Technology / Online Food Delivery

Reason for Selection: Zomato is a food technology platform that connects customers, restaurants, and delivery partners through restaurant discovery, online ordering, payment processing, and delivery management services.

## Step 2: Research

Zomato allows users to discover restaurants, browse menus, place food orders, make payments, track deliveries, and provide ratings and reviews.

### Main Functions:

#### 1. User Management

- Customer registration and login
- Manage customer profiles
- Save delivery addresses

#### 2. Restaurant Discovery

- Search restaurants by location, cuisine, and ratings
- View restaurant details
- Browse food menus

#### 3. Food Ordering

- Add items to cart
- Place orders
- Track order status
- View order history

#### 4. Payment Processing

- Multiple payment methods
- Online transactions
- Payment status tracking

#### 5. Delivery Management

- Assign delivery partners
- Track delivery status

- Monitor delivery time

## 6. Reviews and Ratings

- Customers provide ratings
- Write restaurant reviews
- Analyze customer feedback

### Core Entities Identified

| Entity            | Description                               |
|-------------------|-------------------------------------------|
| Customers         | Stores customer details and profiles      |
| Restaurants       | Stores restaurant information             |
| Locations         | Stores city, area, and delivery locations |
| Menu_Items        | Stores restaurant food items and prices   |
| Orders            | Stores customer order details             |
| Order_Items       | Stores individual items in each order     |
| Payments          | Stores payment transactions               |
| Delivery_Partners | Stores delivery agent information         |
| Deliveries        | Tracks order delivery details             |
| Reviews           | Stores customer ratings and feedback      |

### Step 3: Product Dissection

Zomato improves operational efficiency by using customer, restaurant, order, and delivery data to optimize recommendations, improve customer satisfaction, and support business decision-making.

#### Main Features and Real-World Problems They Solve

| Feature                  | Real-World Problem                                                              | Solution Provided by Zomato                                                             |
|--------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Restaurant Discovery     | Customers find it difficult to search for reliable restaurants and food options | Provides restaurant listings, menus, ratings, reviews, and recommendations in one place |
| Online Food Ordering     | Customers need a convenient way to order food without visiting restaurants      | Enables users to browse menus, add items to cart, and place orders digitally            |
| Real-Time Order Tracking | Customers are uncertain about order status after placing an order               | Provides live tracking and delivery updates                                             |

|                              |                                                           |                                                                            |
|------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------|
| Digital Payments             | Cash payments are inconvenient and time-consuming         | Supports multiple payment options for quick and secure transactions        |
| Customer Reviews & Ratings   | Customers lack information about food quality and service | Helps users make better decisions through ratings and feedback             |
| Delivery Partner Network     | Restaurants need efficient delivery support               | Connects restaurants with delivery partners for faster order fulfillment   |
| Personalized Recommendations | Users spend time searching for suitable food options      | Suggests restaurants and dishes based on preferences and previous activity |

### What Makes Zomato Effective and User-Friendly

- **Easy Restaurant Search:** Users can quickly find restaurants based on location, cuisine, price range, and ratings.
- **Complete Food Ecosystem:** Combines restaurant discovery, ordering, payment, and delivery tracking in a single platform.
- **Data-Driven Recommendations:** Uses customer behavior and preferences to improve food suggestions.
- **Transparent Decision Making:** Ratings, reviews, and photos help customers choose confidently.
- **Convenient Experience:** Reduces the effort and time required to order food.

Product Value Statement : "Zomato solves the problem of discovering, ordering, and receiving food conveniently by connecting customers, restaurants, and delivery partners through a digital food ecosystem."

### Step 4: Case Study on Real-World Problems

#### Challenge 1: Finding Reliable Restaurants

Problem: Customers find it difficult to choose restaurants based on quality, price, and location.

Zomato Solution:

- Provides restaurant listings, menus, ratings, reviews, and filters.
- Helps users compare options and make better decisions.

Example: A customer can filter restaurants by cuisine, rating, and price range to select the best option.

#### Challenge 2: Time-Consuming Food Ordering

Problem: Traditional ordering requires calling restaurants or visiting them.

Zomato Solution:

- Provides online menus, cart, digital ordering, and order tracking.
- Makes food ordering faster and more convenient.

Example: Customers can order meals within minutes from anywhere.

### Challenge 3: Lack of Transparency in Food Quality

Problem: Customers may hesitate to try new restaurants without knowing food quality.

Zomato Solution:

- Uses customer ratings, reviews, and photos to build trust.
- Helps users choose better restaurants.

Example: Users can check reviews before placing an order.

### Step 5: Schema Design Based on Features

| Entity            | Purpose                                   | Attributes                                                                                  |
|-------------------|-------------------------------------------|---------------------------------------------------------------------------------------------|
| Customers         | Stores customer information               | Customer_ID (PK), Name, Email, Phone, Address, Registration_Date                            |
| Restaurants       | Stores restaurant details                 | Restaurant_ID (PK), Restaurant_Name, Location_ID (FK), Cuisine_Type, Rating, Contact_Number |
| Menu_Items        | Stores food items provided by restaurants | Item_ID (PK), Restaurant_ID (FK), Item_Name, Category, Price, Availability_Status           |
| Orders            | Stores customer order details             | Order_ID (PK), Customer_ID (FK), Restaurant_ID (FK), Order_Date, Order_Status, Total_Amount |
| Order_Items       | Stores individual items in each order     | Order_Item_ID (PK), Order_ID (FK), Item_ID (FK), Quantity, Price                            |
| Payments          | Stores payment transaction details        | Payment_ID (PK), Order_ID (FK), Payment_Method, Payment_Status, Transaction_Date            |
| Delivery_Partners | Stores delivery agent details             | Delivery_Partner_ID (PK), Name, Phone, Vehicle_Type                                         |

|                   |                                      |                                                                                                        |
|-------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Deliveries</b> | Tracks order delivery information    | Delivery_ID (PK), Order_ID (FK), Delivery_Partner_ID (FK), Pickup_Time, Delivery_Time, Delivery_Status |
| <b>Reviews</b>    | Stores customer feedback and ratings | Review_ID (PK), Customer_ID (FK), Restaurant_ID (FK), Rating, Review_Text, Review_Date                 |

## Entity Relationships

| Relationship                  | Description                                            |
|-------------------------------|--------------------------------------------------------|
| Customer → Orders             | One customer can place multiple orders (1:M)           |
| Restaurant → Menu_Items       | One restaurant can have multiple menu items (1:M)      |
| Restaurant → Orders           | One restaurant can receive multiple orders (1:M)       |
| Order → Order_Items           | One order can contain multiple food items (1:M)        |
| Order → Payment               | One order has one payment record (1:1)                 |
| Delivery_Partner → Deliveries | One delivery partner can deliver multiple orders (1:M) |
| Customer → Reviews            | One customer can write multiple reviews (1:M)          |
| Restaurant → Reviews          | One restaurant can receive multiple reviews (1:M)      |

## Step 6: Rationale Behind Design

### Database Normalization

The database follows normalization principles by separating entities into different tables. This reduces duplicate data, improves consistency, and makes updates easier.

Example: Instead of storing restaurant details repeatedly in every order:

Bad:

|                                                                    |
|--------------------------------------------------------------------|
| Orders                                                             |
| order_id<br>customer_name<br>restaurant_name<br>restaurant_address |

Good:

|        |
|--------|
| Orders |
|--------|

order\_id  
customer\_id  
restaurant\_id

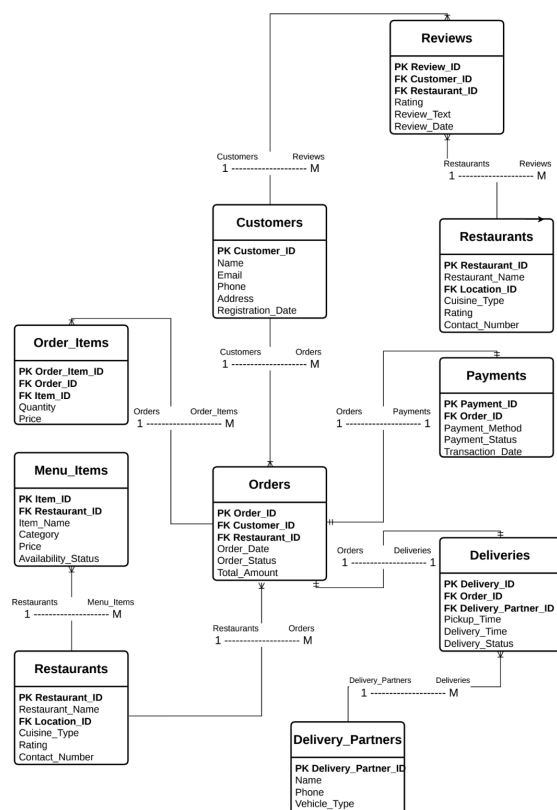
## How the Schema Supports Zomato's Goals

- **Data Organization:** Separates customers, restaurants, orders, payments, and deliveries into structured tables.
- **Reduced Data Duplication:** Normalized design avoids repeating information and improves database efficiency.
- **Better Analytics:** Enables analysis of revenue, customer behavior, restaurant performance, and delivery efficiency.
- **Scalability:** The schema can support increasing users, restaurants, and transactions.
- **Improved Performance:** Primary keys and foreign keys allow faster data retrieval and maintain data consistency.

Overall, this schema supports Zomato's business operations by efficiently managing the complete food delivery lifecycle from restaurant discovery to order completion.

## Step 7: ER Diagram

The ER Diagram represents the relationship between different entities, their attributes, primary keys (PK), and foreign keys (FK) in the Zomato database.



# Relationship Summary

| Relationship                  | Cardinality |
|-------------------------------|-------------|
| Customer → Orders             | One-to-Many |
| Restaurant → Menu_Items       | One-to-Many |
| Restaurant → Orders           | One-to-Many |
| Orders → Order_Items          | One-to-Many |
| Menu_Items → Order_Items      | One-to-Many |
| Orders → Payments             | One-to-One  |
| Orders → Deliveries           | One-to-One  |
| Delivery_Partner → Deliveries | One-to-Many |
| Customer → Reviews            | One-to-Many |
| Restaurant → Reviews          | One-to-Many |

## Step 8: PostgreSQL Implementation

### 1. Create Database

```
postgres=# \l
          List of databases
  Name      | Owner   | Encoding | Locale Provider | Collate | Ctype | Locale | ICU Rules | Access privileges
-----|-----|-----|-----|-----|-----|-----|-----|-----
postgres   | postgres | UTF8     | libc            | C       | C     |         |           | =c/postgres +
template0  | postgres | UTF8     | libc            | C       | C     |         |           | postgres=CtC/postgres
template1  | postgres | UTF8     | libc            | C       | C     |         |           | =c/postgres +
zomato_db  | postgres | UTF8     | libc            | C       | C     |         |           | postgres=CtC/postgres
(4 rows)

postgres=# \c zomato_db
You are now connected to database "zomato_db" as user "postgres".
```

## 2. Create Tables

```
zomato_db=# CREATE TABLE customers (  
zomato_db(#      customer_id SERIAL PRIMARY KEY,  
zomato_db(#      name VARCHAR(100) NOT NULL,  
zomato_db(#      email VARCHAR(100) UNIQUE NOT NULL,  
zomato_db(#      phone VARCHAR(15),  
zomato_db(#      address TEXT,  
zomato_db(#      registration_date DATE DEFAULT CURRENT_DATE  
zomato_db(# );  
CREATE TABLE  
zomato_db=# CREATE TABLE restaurants (  
zomato_db(#      restaurant_id SERIAL PRIMARY KEY,  
zomato_db(#      restaurant_name VARCHAR(150) NOT NULL,  
zomato_db(#      cuisine_type VARCHAR(100),  
zomato_db(#      rating DECIMAL(2,1),  
zomato_db(#      contact_number VARCHAR(15)  
zomato_db(# );  
CREATE TABLE  
zomato_db=# CREATE TABLE menu_items (  
zomato_db(#      item_id SERIAL PRIMARY KEY,  
zomato_db(#      restaurant_id INT NOT NULL,  
zomato_db(#      item_name VARCHAR(100) NOT NULL,  
zomato_db(#      category VARCHAR(50),  
zomato_db(#      price DECIMAL(10,2) NOT NULL,  
zomato_db(#      availability_status BOOLEAN DEFAULT TRUE,  
zomato_db(#  
zomato_db(#      FOREIGN KEY (restaurant_id)  
zomato_db(#      REFERENCES restaurants(restaurant_id)  
zomato_db(# );  
CREATE TABLE
```

```

zomato_db=# CREATE TABLE orders (
zomato_db(#      order_id SERIAL PRIMARY KEY,
zomato_db(#      customer_id INT NOT NULL,
zomato_db(#      restaurant_id INT NOT NULL,
zomato_db(#      order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
zomato_db(#      order_status VARCHAR(50),
zomato_db(#      total_amount DECIMAL(10,2),
zomato_db(#
zomato_db(#      FOREIGN KEY (customer_id)
zomato_db(#      REFERENCES customers(customer_id),
zomato_db(#
zomato_db(#      FOREIGN KEY (restaurant_id)
zomato_db(#      REFERENCES restaurants(restaurant_id)
zomato_db(# );
CREATE TABLE
zomato_db=# CREATE TABLE order_items (
zomato_db(#      order_item_id SERIAL PRIMARY KEY,
zomato_db(#      order_id INT NOT NULL,
zomato_db(#      item_id INT NOT NULL,
zomato_db(#      quantity INT NOT NULL,
zomato_db(#      price DECIMAL(10,2),
zomato_db(#
zomato_db(#      FOREIGN KEY (order_id)
zomato_db(#      REFERENCES orders(order_id),
zomato_db(#
zomato_db(#      FOREIGN KEY (item_id)
zomato_db(#      REFERENCES menu_items(item_id)
zomato_db(# );
CREATE TABLE
zomato_db=# CREATE TABLE payments (
zomato_db(#      payment_id SERIAL PRIMARY KEY,
zomato_db(#      order_id INT UNIQUE NOT NULL,
zomato_db(#      payment_method VARCHAR(50),
zomato_db(#      payment_status VARCHAR(50),
zomato_db(#      transaction_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
zomato_db(#
zomato_db(#      FOREIGN KEY (order_id)
zomato_db(#      REFERENCES orders(order_id)
zomato_db(# );
CREATE TABLE

```

```

zomato_db=# CREATE TABLE delivery_partners (
zomato_db(# delivery_partner_id SERIAL PRIMARY KEY,
zomato_db(# name VARCHAR(100) NOT NULL,
zomato_db(# phone VARCHAR(15),
zomato_db(# vehicle_type VARCHAR(50)
zomato_db(# );
CREATE TABLE
zomato_db=# CREATE TABLE deliveries (
zomato_db(# delivery_id SERIAL PRIMARY KEY,
zomato_db(# order_id INT UNIQUE NOT NULL,
zomato_db(# delivery_partner_id INT NOT NULL,
zomato_db(# pickup_time TIMESTAMP,
zomato_db(# delivery_time TIMESTAMP,
zomato_db(# delivery_status VARCHAR(50),
zomato_db(#
zomato_db(# FOREIGN KEY (order_id)
zomato_db(# REFERENCES orders(order_id),
zomato_db(#
zomato_db(# FOREIGN KEY (delivery_partner_id)
zomato_db(# REFERENCES delivery_partners(delivery_partner_id)
zomato_db(# );
CREATE TABLE
zomato_db=# CREATE TABLE reviews (
zomato_db(# review_id SERIAL PRIMARY KEY,
zomato_db(# customer_id INT NOT NULL,
zomato_db(# restaurant_id INT NOT NULL,
zomato_db(# rating INT CHECK(rating BETWEEN 1 AND 5),
zomato_db(# review_text TEXT,
zomato_db(# review_date DATE DEFAULT CURRENT_DATE,
zomato_db(#
zomato_db(# FOREIGN KEY (customer_id)
zomato_db(# REFERENCES customers(customer_id),
zomato_db(#
zomato_db(# FOREIGN KEY (restaurant_id)
zomato_db(# REFERENCES restaurants(restaurant_id)
zomato_db(# );
CREATE TABLE

```

### 3. List of all Tables (\dt)

```
zomato_db=# \dt
```

#### List of tables

| Schema | Name              | Type  | Owner    |
|--------|-------------------|-------|----------|
| public | customers         | table | postgres |
| public | deliveries        | table | postgres |
| public | delivery_partners | table | postgres |
| public | menu_items        | table | postgres |
| public | order_items       | table | postgres |
| public | orders            | table | postgres |
| public | payments          | table | postgres |
| public | restaurants       | table | postgres |
| public | reviews           | table | postgres |

(9 rows)

## 4. Table structure (\d customers)

```
zomato_db=# \d customers
```

| Column            | Type                   | Collation | Nullable | Default                                        |
|-------------------|------------------------|-----------|----------|------------------------------------------------|
| customer_id       | integer                |           | not null | nextval('customers_customer_id_seq'::regclass) |
| name              | character varying(100) |           | not null |                                                |
| email             | character varying(100) |           | not null |                                                |
| phone             | character varying(15)  |           |          |                                                |
| address           | text                   |           |          |                                                |
| registration_date | date                   |           |          | CURRENT_DATE                                   |

Indexes:  
"customers\_pkey" PRIMARY KEY, btree (customer\_id)  
"customers\_email\_key" UNIQUE CONSTRAINT, btree (email)

Referenced by:  
TABLE "orders" CONSTRAINT "orders\_customer\_id\_fkey" FOREIGN KEY (customer\_id) REFERENCES customers(customer\_id)  
TABLE "reviews" CONSTRAINT "reviews\_customer\_id\_fkey" FOREIGN KEY (customer\_id) REFERENCES customers(customer\_id)

## 5. Table structure (\d orders)

```
zomato_db=# \d orders
```

| Column        | Type                        | Collation | Nullable | Default                                  |
|---------------|-----------------------------|-----------|----------|------------------------------------------|
| order_id      | integer                     |           | not null | nextval('orders_order_id_seq'::regclass) |
| customer_id   | integer                     |           | not null |                                          |
| restaurant_id | integer                     |           | not null |                                          |
| order_date    | timestamp without time zone |           |          | CURRENT_TIMESTAMP                        |
| order_status  | character varying(50)       |           |          |                                          |
| total_amount  | numeric(10,2)               |           |          |                                          |

Indexes:  
"orders\_pkey" PRIMARY KEY, btree (order\_id)

Foreign-key constraints:  
"orders\_customer\_id\_fkey" FOREIGN KEY (customer\_id) REFERENCES customers(customer\_id)  
"orders\_restaurant\_id\_fkey" FOREIGN KEY (restaurant\_id) REFERENCES restaurants(restaurant\_id)

Referenced by:  
TABLE "deliveries" CONSTRAINT "deliveries\_order\_id\_fkey" FOREIGN KEY (order\_id) REFERENCES orders(order\_id)  
TABLE "order\_items" CONSTRAINT "order\_items\_order\_id\_fkey" FOREIGN KEY (order\_id) REFERENCES orders(order\_id)  
TABLE "payments" CONSTRAINT "payments\_order\_id\_fkey" FOREIGN KEY (order\_id) REFERENCES orders(order\_id)

## 5. Insert Data into Table

```
zomato_db=# INSERT INTO customers
zomato_db=# (name, email, phone, address)
zomato_db=# VALUES
zomato_db=# ('Rahul Sharma','rahul@gmail.com','9876543210','Delhi'),
zomato_db=# ('Priya Verma','priya@gmail.com','9876543211','Mumbai'),
zomato_db=# ('Amit Kumar','amit@gmail.com','9876543212','Bangalore'),
zomato_db=# ('Sneha Patel','sneha@gmail.com','9876543213','Ahmedabad'),
[zomato_db=# ('Arjun Singh','arjun@gmail.com','9876543214','Pune');
INSERT 0 5
zomato_db=# INSERT INTO restaurants
zomato_db=# (restaurant_name, cuisine_type, rating, contact_number)
zomato_db=# VALUES
zomato_db=# ('Biryani House','Indian',4.5,'9999999991'),
zomato_db=# ('Pizza Corner','Italian',4.2,'9999999992'),
zomato_db=# ('South Spice','South Indian',4.7,'9999999993'),
zomato_db=# ('Burger Point','Fast Food',4.0,'9999999994'),
[zomato_db=# ('Healthy Bowl','Healthy Food',4.6,'9999999995');
INSERT 0 5
zomato_db=# INSERT INTO restaurants
zomato_db=# (restaurant_name, cuisine_type, rating, contact_number)
zomato_db=# VALUES
zomato_db=# ('Biryani House','Indian',4.5,'9999999991'),
zomato_db=# ('Pizza Corner','Italian',4.2,'9999999992'),
zomato_db=# ('South Spice','South Indian',4.7,'9999999993'),
zomato_db=# ('Burger Point','Fast Food',4.0,'9999999994'),
[zomato_db=# ('Healthy Bowl','Healthy Food',4.6,'9999999995');
INSERT 0 5
```

```

zomato_db=# INSERT INTO menu_items
zomato_db=# (restaurant_id, item_name, category, price)
zomato_db=# VALUES
zomato_db=# (1, 'Chicken Biryani', 'Main Course', 250),
zomato_db=# (1, 'Veg Biryani', 'Main Course', 180),
zomato_db=# (2, 'Farmhouse Pizza', 'Pizza', 450),
zomato_db=# (2, 'Cheese Pizza', 'Pizza', 350),
zomato_db=# (3, 'Masala Dosa', 'South Indian', 120),
zomato_db=# (3, 'Idli Sambar', 'Breakfast', 80),
zomato_db=# (4, 'Veg Burger', 'Fast Food', 150),
zomato_db=# (4, 'French Fries', 'Snacks', 100),
zomato_db=# (5, 'Protein Bowl', 'Healthy', 300),
[zomato_db=# (5, 'Salad Bowl', 'Healthy', 220);
INSERT 0 10
zomato_db=# INSERT INTO delivery_partners
zomato_db=# (name, phone, vehicle_type)
zomato_db=# VALUES
zomato_db=# ('Ramesh Kumar', '9000000001', 'Bike'),
zomato_db=# ('Suresh Patel', '9000000002', 'Scooter'),
zomato_db=# ('Vijay Singh', '9000000003', 'Bike'),
[zomato_db=# ('Karan Das', '9000000004', 'Cycle');
INSERT 0 4
zomato_db=# INSERT INTO orders
zomato_db=# (customer_id, restaurant_id, order_status, total_amount)
zomato_db=# VALUES
zomato_db=# (1, 1, 'Delivered', 430),
zomato_db=# (2, 2, 'Delivered', 450),
zomato_db=# (3, 3, 'Delivered', 200),
zomato_db=# (4, 4, 'Preparing', 250),
zomato_db=# (5, 5, 'Out for Delivery', 520),
zomato_db=# (1, 3, 'Delivered', 300),
[zomato_db=# (2, 1, 'Cancelled', 250);
INSERT 0 7

```

```

zomato_db=# INSERT INTO order_items
zomato_db=# (order_id, item_id, quantity, price)
zomato_db=# VALUES
zomato_db=# (1,1,1,250),
zomato_db=# (1,2,1,180),
zomato_db=#
zomato_db=# (2,3,1,450),
zomato_db=#
zomato_db=# (3,5,1,120),
zomato_db=# (3,6,1,80),
zomato_db=#
zomato_db=# (4,7,1,150),
zomato_db=# (4,8,1,100),
zomato_db=#
zomato_db=# (5,9,1,300),
zomato_db=# (5,10,1,220),
zomato_db=#
zomato_db=# (6,5,2,240),
zomato_db=#
zomato_db=# (7,1,1,250);
INSERT 0 11
zomato_db=# INSERT INTO payments
zomato_db=# (order_id, payment_method, payment_status)
zomato_db=# VALUES
zomato_db=# (1,'UPI','Completed'),
zomato_db=# (2,'Credit Card','Completed'),
zomato_db=# (3,'UPI','Completed'),
zomato_db=# (4,'Cash on Delivery','Pending'),
zomato_db=# (5,'UPI','Completed'),
zomato_db=# (6,'Debit Card','Completed'),
zomato_db=# (7,'UPI','Refunded');
INSERT 0 7
zomato_db=# INSERT INTO deliveries
zomato_db=# (order_id, delivery_partner_id, pickup_time, delivery_time, delivery_status)
zomato_db=# VALUES
zomato_db=# (1,1,'2026-06-17 12:00','2026-06-17 12:40','Delivered'),
zomato_db=# (2,2,'2026-06-17 13:00','2026-06-17 13:45','Delivered'),
zomato_db=# (3,3,'2026-06-17 14:00','2026-06-17 14:30','Delivered'),
zomato_db=# (4,4,'2026-06-17 15:00',NULL,'Preparing'),
zomato_db=# (5,1,'2026-06-17 16:00',NULL,'Out for Delivery'),
zomato_db=# (6,2,'2026-06-17 17:00','2026-06-17 17:40','Delivered');
INSERT 0 6
zomato_db=# INSERT INTO reviews
zomato_db=# (customer_id, restaurant_id, rating, review_text)
zomato_db=# VALUES
zomato_db=# (1,1,5,'Excellent biryani and fast delivery'),
zomato_db=# (2,2,4,'Good pizza quality'),
zomato_db=# (3,3,5,'Best South Indian food'),
zomato_db=# (4,4,3,'Average burger'),
zomato_db=# (5,5,5,'Healthy and tasty food'),
zomato_db=# (1,3,4,'Good taste and service');
INSERT 0 6

```

## 6. Display the data

### a. Customers

```

zomato_db=# SELECT * FROM customers;

```

| customer_id | name         | email           | phone      | address   | registration_date |
|-------------|--------------|-----------------|------------|-----------|-------------------|
| 1           | Rahul Sharma | rahul@gmail.com | 9876543210 | Delhi     | 2026-06-18        |
| 2           | Priya Verma  | priya@gmail.com | 9876543211 | Mumbai    | 2026-06-18        |
| 3           | Amit Kumar   | amit@gmail.com  | 9876543212 | Bangalore | 2026-06-18        |
| 4           | Sneha Patel  | sneha@gmail.com | 9876543213 | Ahmedabad | 2026-06-18        |
| 5           | Arjun Singh  | arjun@gmail.com | 9876543214 | Pune      | 2026-06-18        |

(5 rows)

## b. Restaurants

```
zomato_db=# SELECT * FROM restaurants;
 restaurant_id | restaurant_name | cuisine_type | rating | contact_number
-----|-----|-----|-----|-----
          1 | Biryani House | Indian | 4.5 | 9999999991
          2 | Pizza Corner | Italian | 4.2 | 9999999992
          3 | South Spice | South Indian | 4.7 | 9999999993
          4 | Burger Point | Fast Food | 4.0 | 9999999994
          5 | Healthy Bowl | Healthy Food | 4.6 | 9999999995
          6 | Biryani House | Indian | 4.5 | 9999999991
          7 | Pizza Corner | Italian | 4.2 | 9999999992
          8 | South Spice | South Indian | 4.7 | 9999999993
          9 | Burger Point | Fast Food | 4.0 | 9999999994
         10 | Healthy Bowl | Healthy Food | 4.6 | 9999999995
(10 rows)
```

## c. Orders

```
zomato_db=# SELECT * FROM orders;
 order_id | customer_id | restaurant_id | order_date | order_status | total_amount
-----|-----|-----|-----|-----|-----
          1 |          1 |          1 | 2026-06-18 11:02:46.381687 | Delivered | 430.00
          2 |          2 |          2 | 2026-06-18 11:02:46.381687 | Delivered | 450.00
          3 |          3 |          3 | 2026-06-18 11:02:46.381687 | Delivered | 200.00
          4 |          4 |          4 | 2026-06-18 11:02:46.381687 | Preparing | 250.00
          5 |          5 |          5 | 2026-06-18 11:02:46.381687 | Out for Delivery | 520.00
          6 |          1 |          3 | 2026-06-18 11:02:46.381687 | Delivered | 300.00
          7 |          2 |          1 | 2026-06-18 11:02:46.381687 | Cancelled | 250.00
(7 rows)
```

## d. Payments

```
zomato_db=#
zomato_db=# SELECT * FROM payments;
 payment_id | order_id | payment_method | payment_status | transaction_date
-----|-----|-----|-----|-----
          1 |          1 | UPI | Completed | 2026-06-18 11:03:02.540898
          2 |          2 | Credit Card | Completed | 2026-06-18 11:03:02.540898
          3 |          3 | UPI | Completed | 2026-06-18 11:03:02.540898
          4 |          4 | Cash on Delivery | Pending | 2026-06-18 11:03:02.540898
          5 |          5 | UPI | Completed | 2026-06-18 11:03:02.540898
          6 |          6 | Debit Card | Completed | 2026-06-18 11:03:02.540898
          7 |          7 | UPI | Refunded | 2026-06-18 11:03:02.540898
(7 rows)
```

## d. Reviews

```
zomato_db=# SELECT * FROM reviews;
 review_id | customer_id | restaurant_id | rating | review_text | review_date
-----|-----|-----|-----|-----|-----
          1 |          1 |          1 | 5 | Excellent biryani and fast delivery | 2026-06-18
          2 |          2 |          2 | 4 | Good pizza quality | 2026-06-18
          3 |          3 |          3 | 5 | Best South Indian food | 2026-06-18
          4 |          4 |          4 | 3 | Average burger | 2026-06-18
          5 |          5 |          5 | 5 | Healthy and tasty food | 2026-06-18
          6 |          1 |          3 | 4 | Good taste and service | 2026-06-18
(6 rows)
```

## Step 9: SQL Analysis Queries

### 1. Top Restaurants by Number of Orders

Business Question: Which restaurants receive the highest number of orders?

Insight: Helps identify high-performing restaurants and popular food partners.

```
zomato_db=# SELECT
zomato_db=#     r.restaurant_name,
zomato_db=#     COUNT(o.order_id) AS total_orders
zomato_db=# FROM restaurants r
zomato_db=# JOIN orders o
zomato_db=# ON r.restaurant_id = o.restaurant_id
zomato_db=# GROUP BY r.restaurant_name
zomato_db=# ORDER BY total_orders DESC;
 restaurant_name | total_orders
-----+-----
South Spice      |           2
Biryani House    |           2
Healthy Bowl     |           1
Burger Point     |           1
Pizza Corner     |           1
(5 rows)
```

## 2. Total Revenue Generated by Each Restaurant

Business Question: Which restaurants generate the highest revenue?

Insight: Identifies restaurants contributing maximum revenue.

```
zomato_db=# SELECT
zomato_db=#     r.restaurant_name,
zomato_db=#     SUM(o.total_amount) AS total_revenue
zomato_db=# FROM restaurants r
zomato_db=# JOIN orders o
zomato_db=# ON r.restaurant_id = o.restaurant_id
zomato_db=# WHERE o.order_status = 'Delivered'
zomato_db=# GROUP BY r.restaurant_name
zomato_db=# ORDER BY total_revenue DESC;
 restaurant_name | total_revenue
-----+-----
 South Spice      |          500.00
 Pizza Corner     |          450.00
 Biryani House    |          430.00
(3 rows)
```

## 3. Customer Spending Analysis

Business Question: Who are the highest spending customers?

Insight: Helps identify valuable customers for loyalty programs.

```
zomato_db=# SELECT
zomato_db=#     c.name AS customer_name,
zomato_db=#     COUNT(o.order_id) AS total_orders,
zomato_db=#     SUM(o.total_amount) AS total_spent
zomato_db=# FROM customers c
zomato_db=# JOIN orders o
zomato_db=# ON c.customer_id = o.customer_id
zomato_db=# WHERE o.order_status = 'Delivered'
zomato_db=# GROUP BY c.name
zomato_db=# ORDER BY total_spent DESC;
 customer_name | total_orders | total_spent
-----+-----+-----
 Rahul Sharma  |           2 |       730.00
 Priya Verma   |           1 |       450.00
 Amit Kumar    |           1 |       200.00
(3 rows)
```

## 4. Restaurant Average Rating Analysis

Business Question: Which restaurants have the best customer ratings?

Insight: Shows customer satisfaction and restaurant quality.

```
zomato_db=# SELECT
zomato_db=#     r.restaurant_name,
zomato_db=#     ROUND(AVG(rv.rating),2) AS average_rating,
zomato_db=#     COUNT(rv.review_id) AS total_reviews
zomato_db=# FROM restaurants r
zomato_db=# JOIN reviews rv
zomato_db=# ON r.restaurant_id = rv.restaurant_id
zomato_db=# GROUP BY r.restaurant_name
zomato_db=# ORDER BY average_rating DESC;
 restaurant_name | average_rating | total_reviews
-----+-----+-----
Healthy Bowl    |          5.00 |            1
Biryani House   |          5.00 |            1
South Spice     |          4.50 |            2
Pizza Corner    |          4.00 |            1
Burger Point    |          3.00 |            1
(5 rows)
```

## 5. Order Status Analysis

Business Question: What percentage of orders are completed, cancelled, or pending?

Insight: Helps analyze operational efficiency.

```
zomato_db=# SELECT
zomato_db=#     order_status,
zomato_db=#     COUNT(order_id) AS total_orders
zomato_db=# FROM orders
zomato_db=# GROUP BY order_status
zomato_db=# ORDER BY total_orders DESC;
 order_status | total_orders
-----+-----
Delivered    |            4
Preparing    |            1
Out for Delivery |            1
Cancelled    |            1
(4 rows)
```

## 6. Most Popular Food Items

Business Question: Which menu items are ordered the most?

Insight: Helps restaurants optimize menu planning and inventory.

```
zomato_db=# SELECT
zomato_db=#     m.item_name,
zomato_db=#     SUM(oi.quantity) AS total_quantity_sold
zomato_db=# FROM menu_items m
zomato_db=# JOIN order_items oi
zomato_db=# ON m.item_id = oi.item_id
zomato_db=# GROUP BY m.item_name
zomato_db=# ORDER BY total_quantity_sold DESC;
```

| item_name       | total_quantity_sold |
|-----------------|---------------------|
| Masala Dosa     | 3                   |
| Chicken Biryani | 2                   |
| Veg Burger      | 1                   |
| French Fries    | 1                   |
| Salad Bowl      | 1                   |
| Veg Biryani     | 1                   |
| Farmhouse Pizza | 1                   |
| Idli Sambar     | 1                   |
| Protein Bowl    | 1                   |

(9 rows)

## 7. Delivery Performance Analysis

Business Question: How many deliveries are completed by each delivery partner?

Insight: Measures delivery partner performance.

```
zomato_db=# SELECT
zomato_db=#     dp.name AS delivery_partner,
zomato_db=#     COUNT(d.delivery_id) AS completed_deliveries
zomato_db=# FROM delivery_partners dp
zomato_db=# JOIN deliveries d
zomato_db=# ON dp.delivery_partner_id = d.delivery_partner_id
zomato_db=# WHERE d.delivery_status = 'Delivered'
zomato_db=# GROUP BY dp.name
zomato_db=# ORDER BY completed_deliveries DESC;
```

| delivery_partner | completed_deliveries |
|------------------|----------------------|
| Suresh Patel     | 2                    |
| Ramesh Kumar     | 1                    |
| Vijay Singh      | 1                    |

(3 rows)

## 8. Customer Segmentation Using CASE

Business Question: Classify customers based on their spending behavior.

Insight: Supports targeted marketing campaigns.

```
zomato_db=# SELECT
zomato_db=#     c.name,
zomato_db=#     SUM(o.total_amount) AS total_spent,
zomato_db=#     CASE
zomato_db=#         WHEN SUM(o.total_amount) >= 500 THEN 'High Value Customer'
zomato_db=#         WHEN SUM(o.total_amount) >= 250 THEN 'Medium Value Customer'
zomato_db=#         ELSE 'Low Value Customer'
zomato_db=#     END AS customer_segment
zomato_db=# FROM customers c
zomato_db=# JOIN orders o
zomato_db=# ON c.customer_id = o.customer_id
zomato_db=# WHERE o.order_status='Delivered'
zomato_db=# GROUP BY c.name
zomato_db=# ORDER BY total_spent DESC;
```

| name         | total_spent | customer_segment      |
|--------------|-------------|-----------------------|
| Rahul Sharma | 730.00      | High Value Customer   |
| Priya Verma  | 450.00      | Medium Value Customer |
| Amit Kumar   | 200.00      | Low Value Customer    |

(3 rows)

## 9. Rank Restaurants by Revenue (Window Function)

Business Question: What is the revenue ranking of restaurants?

Insight: Ranks restaurants based on business performance.

```
zomato_db=# SELECT
zomato_db=#     restaurant_name,
zomato_db=#     total_revenue,
zomato_db=#     RANK() OVER(
zomato_db=#         ORDER BY total_revenue DESC
zomato_db=#     ) AS revenue_rank
zomato_db=# FROM
zomato_db=# (
zomato_db=#     SELECT
zomato_db=#         r.restaurant_name,
zomato_db=#         SUM(o.total_amount) AS total_revenue
zomato_db=#     FROM restaurants r
zomato_db=#     JOIN orders o
zomato_db=#     ON r.restaurant_id=o.restaurant_id
zomato_db=#     WHERE o.order_status='Delivered'
zomato_db=#     GROUP BY r.restaurant_name
zomato_db=# ) revenue_table;
 restaurant_name | total_revenue | revenue_rank
-----+-----+-----
South Spice      |      500.00   |           1
Pizza Corner     |      450.00   |           2
Biryani House    |      430.00   |           3
(3 rows)
```

## 10. Monthly Revenue Trend (CTE)

Business Question: How does revenue change over time?

Insight: Helps identify growth trends and seasonal demand.

```
zomato_db=# WITH monthly_sales AS
zomato_db=# (
zomato_db=#     SELECT
zomato_db=#         DATE_TRUNC('month', order_date) AS month,
zomato_db=#         SUM(total_amount) AS revenue
zomato_db=#     FROM orders
zomato_db=#     WHERE order_status='Delivered'
zomato_db=#     GROUP BY month
zomato_db=# )
zomato_db=#
zomato_db=# SELECT *
zomato_db=# FROM monthly_sales
zomato_db=# ORDER BY month;
      month      | revenue
-----+-----
2026-06-01 00:00:00 | 1380.00
(1 row)
```

### Summary of SQL Query (Skills) Demonstrated

| Skill               | Query                         |
|---------------------|-------------------------------|
| INNER JOIN          | Restaurant & Order Analysis   |
| GROUP BY            | Revenue and Customer Analysis |
| Aggregate Functions | SUM, COUNT, AVG               |
| CASE Statement      | Customer Segmentation         |
| Window Function     | Restaurant Ranking            |
| CTE                 | Monthly Revenue Trend         |
| Business Analysis   | Insights Generation           |

## Step 10: Key Insights & Conclusion

### Key Insights

- **Restaurant Performance:** Identified top restaurants based on orders, revenue, and customer ratings.
- **Customer Analysis:** Segmented customers based on spending behavior to support personalized offers.
- **Revenue Analysis:** Analyzed sales trends and identified high-performing restaurants and food items.
- **Menu Analysis:** Found popular dishes to improve inventory and menu planning.
- **Delivery Analysis:** Evaluated delivery performance to improve operational efficiency.
- **Customer Satisfaction:** Used ratings and reviews to measure service quality.

### Conclusion

The Zomato SQL project demonstrates how a well-designed relational database helps manage customers, restaurants, orders, payments, deliveries, and reviews.

Using SQL analysis, the platform can gain business insights, improve customer experience, optimize operations, and support data-driven decision-making.